

TANMAY KUMAR

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EDUCATION

Computer Science Engineering | SRM Institute of Science and Technology,
Kattankulathur

CGPA: 9.80 | 2024-2028

TECHNICAL SKILLS

- Programming Languages: Java, C++, Python, C
- Web Technologies: HTML, CSS, React, TypeScript, Tailwind CSS, Framer Motion
- Databases: MySQL, PostgreSQL, MongoDB, Redis
- Tools & Platforms: Maven, Power BI, Android Studio, Blender, Unity, Vite, Node.js, FastAPI

PROJECTS

CircuitVerse [Link](#)

React + TypeScript, Vite, Tailwind CSS, Framer Motion, WebXR, Model-Viewer, HLS.js, Blender

- College-backed AR/VR web platform where students scan QR codes to explore 3D circuit models with interactive component hotspots, wiring details, and quizzes.
- Built a Liquid-Glass UI with Framer Motion and HLS streaming; integrated WebXR and Google Model-Viewer for real-world AR circuit placement.

Project Lifeline [Link](#)

HTML5, JavaScript, Tailwind CSS, Vis-Network

- Browser-based algorithm visualization dashboard across three phases: Fractional Knapsack, Floyd-Warshall shortest paths with a live interactive graph, and Subset Sum with a dynamic State Space Tree.
- Destroying routes triggers live $O(V^3)$ matrix recalculation; an Efficiency Index Monitor logs complexity and performance across all phases in real time.

GigSecure – AI-Powered Parametric Income Protection

React + Vite (PWA), Node.js + Express, PostgreSQL, MongoDB, Redis, Python Flask, scikit-learn, Socket.IO

- Parametric insurance platform for Q-Commerce gig riders that monitors external APIs and auto-triggers instant payouts on disruptions — zero claim filing required.
- ML pipeline using XGBoost for risk profiling, ARIMA for claim forecasting, and Isolation Forest for GPS spoofing detection; real-time alerts via Socket.IO.

GENGO – AI-Powered Natural Language Database Analyst

Python, FastAPI, Streamlit, PostgreSQL (Supabase), Groq/Llama-3, Gemini 2.0 Flash

- Translates plain English to optimized PostgreSQL queries using LLMs with live schema awareness; blocks all destructive SQL commands via a hardcoded read-only safety layer.
- Self-healing loop feeds SQL exceptions back to the LLM for autonomous correction; Streamlit frontend auto-charts results with CSV export, deployed on Heroku + Streamlit Cloud.

ACHIEVEMENTS

- Top 25 nationally in a Namespace-organized hackathon (2025), competing among teams across India.
- Secured First Place at the Poster Presentation – SUMMIT VC 2.0 hosted by SRM Institute of Science and Technology; presented CircuitVerse, an AR-based 3D visualization platform for making circuit understanding more interactive and accessible.

LEADERSHIP & ACTIVITIES

- Contributed as a volunteer for the Global Outreach Workshop on “Building Smart Solutions with IoT” conducted at the Spatial Computing Lab, SRMIST – an insightful session on IoT and smart systems.
- Member, Futurix Club (Public Relations): Assisted in organizing multiple technical and non-technical events.